INCH-POUND

MIL-T-27/74B
4 May 2012
SUPERSEDING
MIL-T-27/74A
13 March 1991

MILITARY SPECIFICATION SHEET

TRANSFORMER, AUDIO FREQUENCY

Inactive for new design after 13 March 1991

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the products described herein shall consist of this specification and MIL-PRF-27.

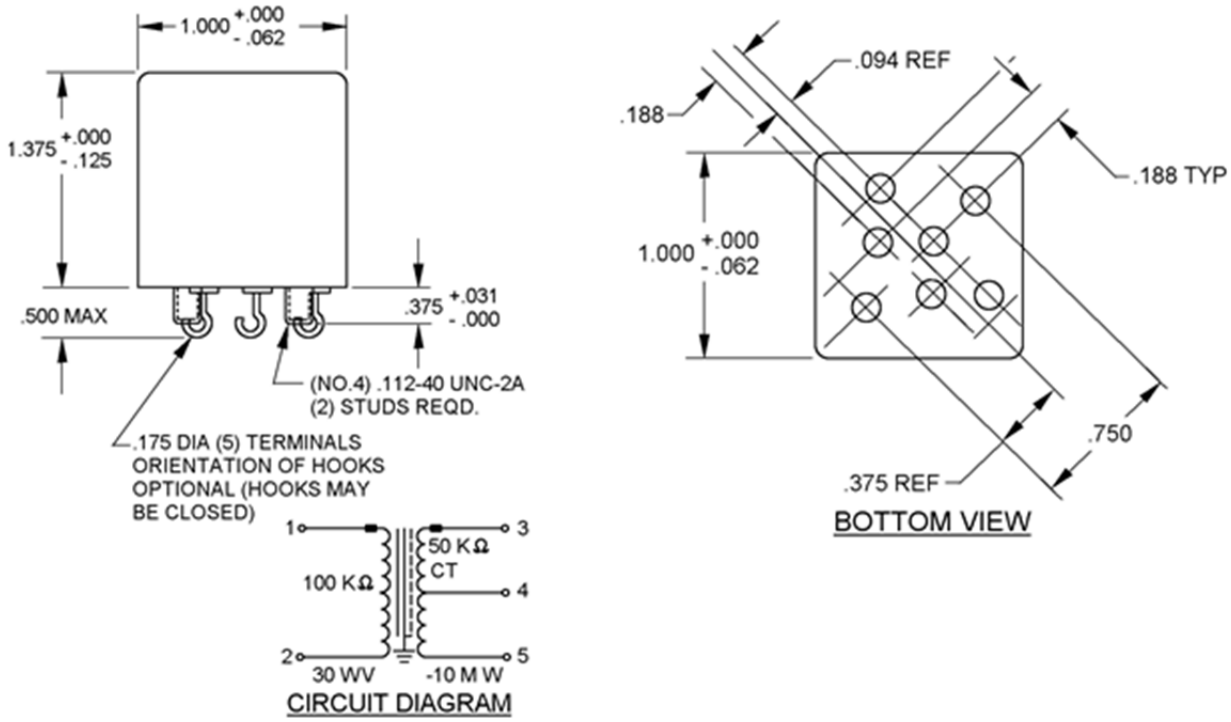


FIGURE 1. Dimensions and configuration

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Inches	mm	Inches	mm
.000	0.00	.188	4.78
.031	0.79	.375	9.52
.062	1.57	.500	12.70
.094	2.39	.750	19.05
.112	2.84	1.000	25.40
.125	3.18	1.375	34.92

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.016 (0.41 mm).
4. Marking shall be on sides or top of case.

FIGURE 1. Dimensions and configuration – Continued.

REQUIREMENTS: (When numbers in parentheses, i.e., (1-3) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings: See table I.

Primary power level: 10 mW maximum.

Working voltage : 30 volts.

Primary current: Zero.

Primary impedance (1-2): 100 kilohms.

Design and construction:

Dimensions and configuration: See figure 1.

Duty cycle: Continuous.

Altitude: 10,000 feet.

Terminals: Wire hook.

Diameter: 0.175 inch.

Length: 0.500 inch maximum.

Weight: 0.2 pound.

Operating temperature range: -55°C to +105°C.

Terminal strength: MIL-STD-202, method 211, test condition A, 2 pounds.

Dielectric withstanding voltage (at sea level): 500 V rms.

Electrical characteristics: See table II.

No load (centertap voltage unbalance only): With 5 V, 1 kHz applied to (1-2); unbalance (3-4) and (4-5) ±1 percent maximum.

Harmonic distortion: Total harmonic content of the output: 5% with 5 V, 400 Hz applied to (1-2) maximum.

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Primary impedance: 100 kilohms with approximately 5 V, 400 Hz applied to (1-2) and 50 kilohms across (3-5).

Frequency response:

Z_S = 100 kilohms (1-2)

Z_L = 50 kilohms (3-5)

E_S = 5 V

Reference frequency = 1 kHz.

Frequency range is 100 to 10,000 Hz ±2 dB.

Insertion loss: 1 dB with 10 mW, 400 Hz across (1-2) maximum..

Resonance: Second resonant frequency – 20 kHz minimum.

Electrostatic shielding: Voltage ratio, 5:1 minimum at 20 kHz.

Polarity: Additive, with terminals 2 and 3 connected.

VERIFICATION

Qualification inspection: Not applicable.

Conformance inspection: Groups A and B tests of MIL-PRF-27 shall be applicable.

Marking location: See figure 1.

Part or Identifying Number (PIN): M27/74-01.

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents. In addition to MIL-PRF-27, this document references the following:

MIL-STD-202

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA – CC

(Project 5950-2009-073)

Review activities:
Army - AR, CR4, MI
Navy - AS, MC, OS, SH
Air Force - 19

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of this information above using the ASSIST Online database at https://assist.daps.dla.mil.

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